Machine Learning CS 4641

# Information Theory

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*Slides adapted from Max Roozbahani, Le Song , Roni Rosenfeld, Maneesh Sahani, and Chao Zhang.*

# Outline

- Motivation
- Entropy
- Conditional Entropy and Mutual Information
- Cross-Entropy and KL-Divergence



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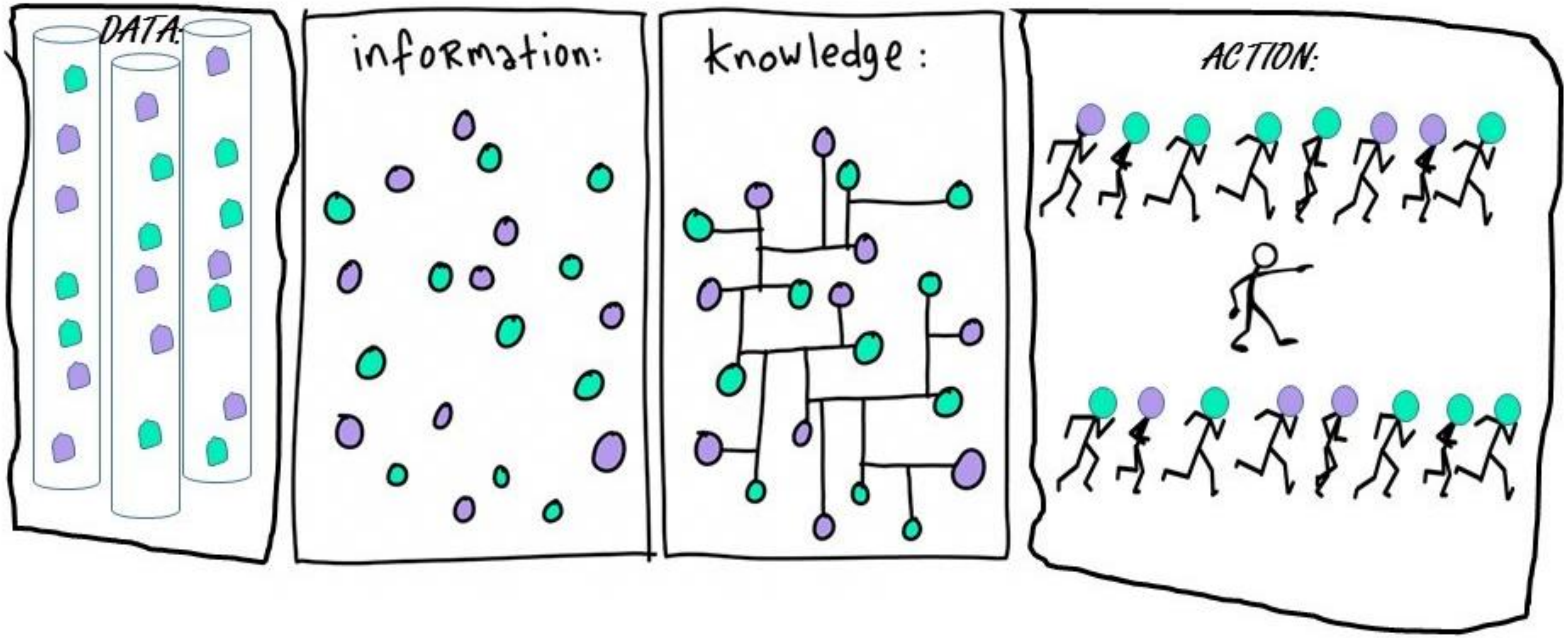
# Uncertainty and Information

**Information** is processed data whereas **knowledge** is **information** that is modeled to be useful.

You need **information** to be able to get **knowledge**

- **More information when an unlikely event occurs than when something certain occurs** (in fact, it should be zero when the event is certain)
- **Example:** You are in sunny Los Angeles, California and you are told it did not rain yesterday → **not a lot of information since it rarely rains in SoCal**

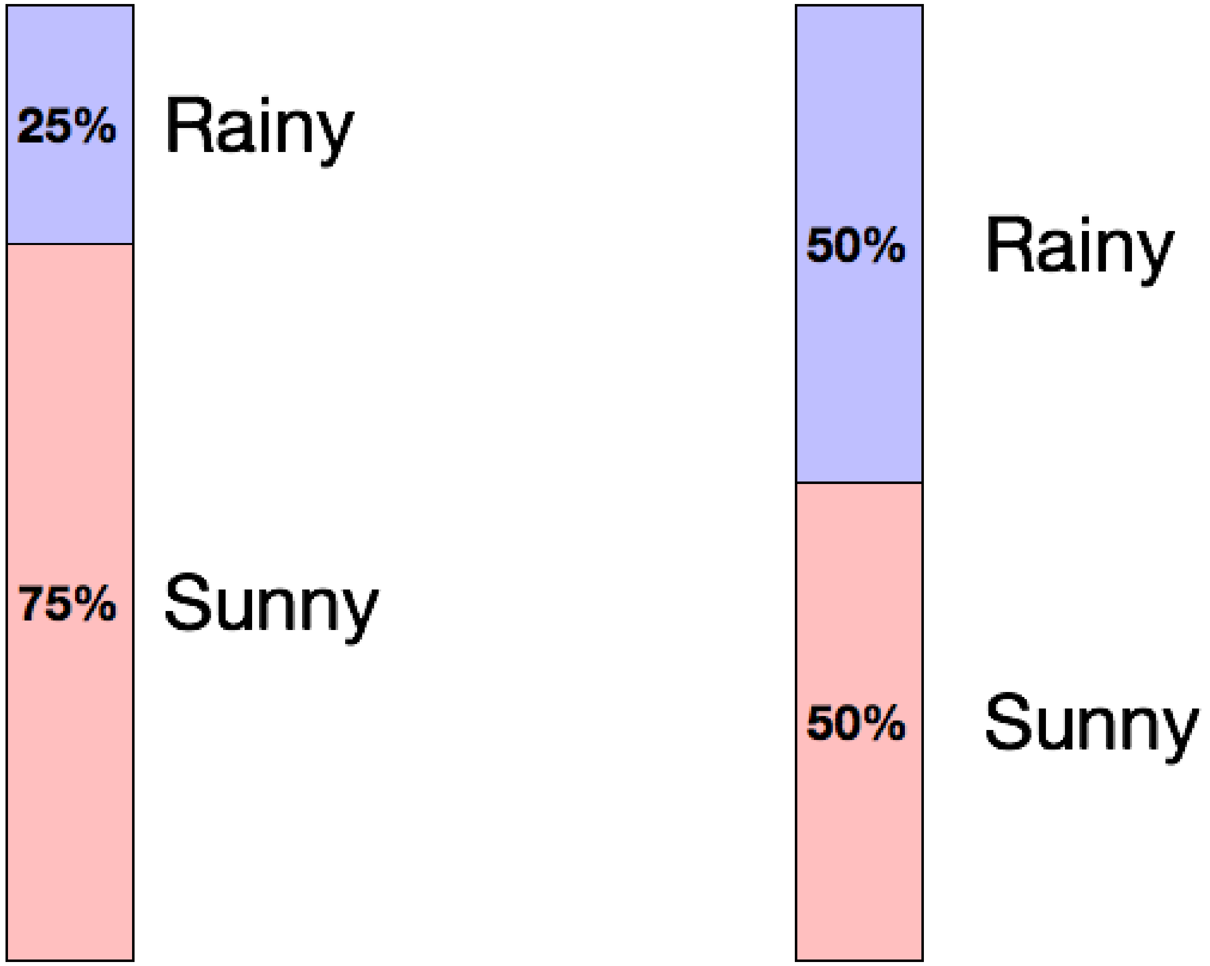



Created by Bruce Campbell: “DIKA – ancient Chinese saying for get up and DO! Data-Information-Knowledge-Action.”

# Information Theory

- Information theory is a mathematical framework which addresses questions like:
  - How much information does a random variable carry about?
  - How efficient is a hypothetical code, given the statistics of the random variable?
  - How can we encode more efficiently?
  - Is the **information carried by different random variables complementary or redundant**?

# Uncertainty and Information




**Which day is more uncertain?**

**How do we quantify uncertainty?**

Less likely → More uncertainty → More information → High entropy

# Information

is always +ve

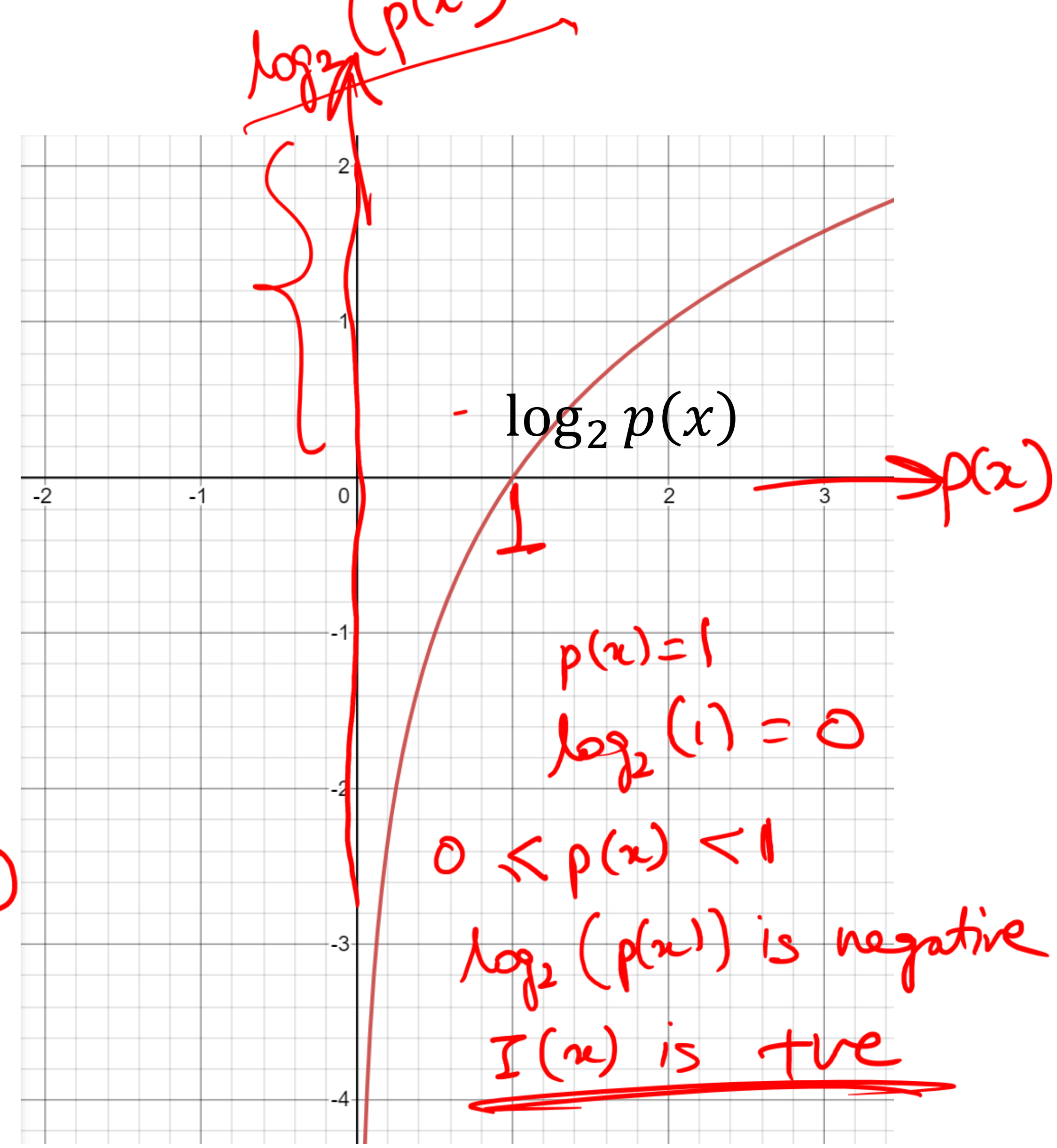
We can associate our measure of information with probability of an event occurring. Let X be a random variable with distribution $p(x)$:

$$I(x) = \boxed{\log_2 \frac{1}{p(x)}} = \boxed{-\log_2 p(x)}$$

is in bits

$\log_2(p(x))^{-1} = -1 * \log_2 p(x)$

High probability means less information



# Information: Example

$p(x=\text{dog}) < p(x=\text{cat})$

$I(x=\text{dog}) > I(x=\text{cat})$

X = [cat, cat, cat, dog ]

$p(x=\text{cat}) = 1$

$p(x=\text{cat}) = 1$

$p(x=\text{cat}) = 1$

$p(x=\text{cat}) = 3/4 = 0.75$

$p(x=\text{dog}) = 1/4 = 0.25$

$$I(x=\text{cat}) = \log_2 \frac{1}{p(x=\text{cat})} = \log_2 \frac{1}{3/4} = \log_2 4/3$$

$$I(x=\text{dog}) = \log_2 \frac{1}{p(x=\text{dog})} = \log_2 \frac{1}{1/4} = \log_2 4 > \log_2 4/3$$

# Example: is a picture worth 1,000 words?

- Information obtained by a random word from a 100,000 word vocabulary:

$$I(word) = \log_2\left(\frac{1}{p(x)}\right) = \log_2\left(\frac{1}{1/100{,}000}\right) = 16.61\ bits$$

- A 1,000-word document from same source:

$$I(document) = 1000 \times I(word) = 16{,}610\ bits$$

- A 640 x 480 pixel, 16-greyscale picture (each pixel has 16 bits information):

$$I(picture) = \log_2\left(\frac{1}{1/16^{640\times480}}\right) = 1{,}228{,}800\ bits$$

**A picture is worth (a lot more than) 1,000 words!**

# Motivation: Data compression

- Suppose we observe a sequence of events
  - Coin tosses
  - Words in a language
  - Notes in a song
  - etc.
- We want to record the sequence of events in the smallest possible space
- In other words, we want the shortest representation which preserves the information
- Another way to think about this: **how much information does the sequence of events actually contain?**

# Example: Data compression

- Consider the problem of recording coin tosses in unary

T, T, T, T, H

Sample space

- **Approach 1:**

Requirement

0s & 00s

| Heads | Tails |
|---|---|
| 0 | 00 |

00, 00, 00, 00, 0

We used **9** characters

- Which one has a higher probability: T or H?
- Which one should carry more information: T or H?

# Example: Data compression

$p(H) = 1/5$ ← should be encoded with high Information
$p(T) = 4/5$

- Consider the problem of recording coin tosses in unary

T, T, T, T, H

$H \to 00$
$T \to 0$

if $p(x) \uparrow$, $I \downarrow$

- **Approach 2:**

| Heads | Tails |
|---|---|
| 00 | 0 |

0, 0, 0, 0, 00

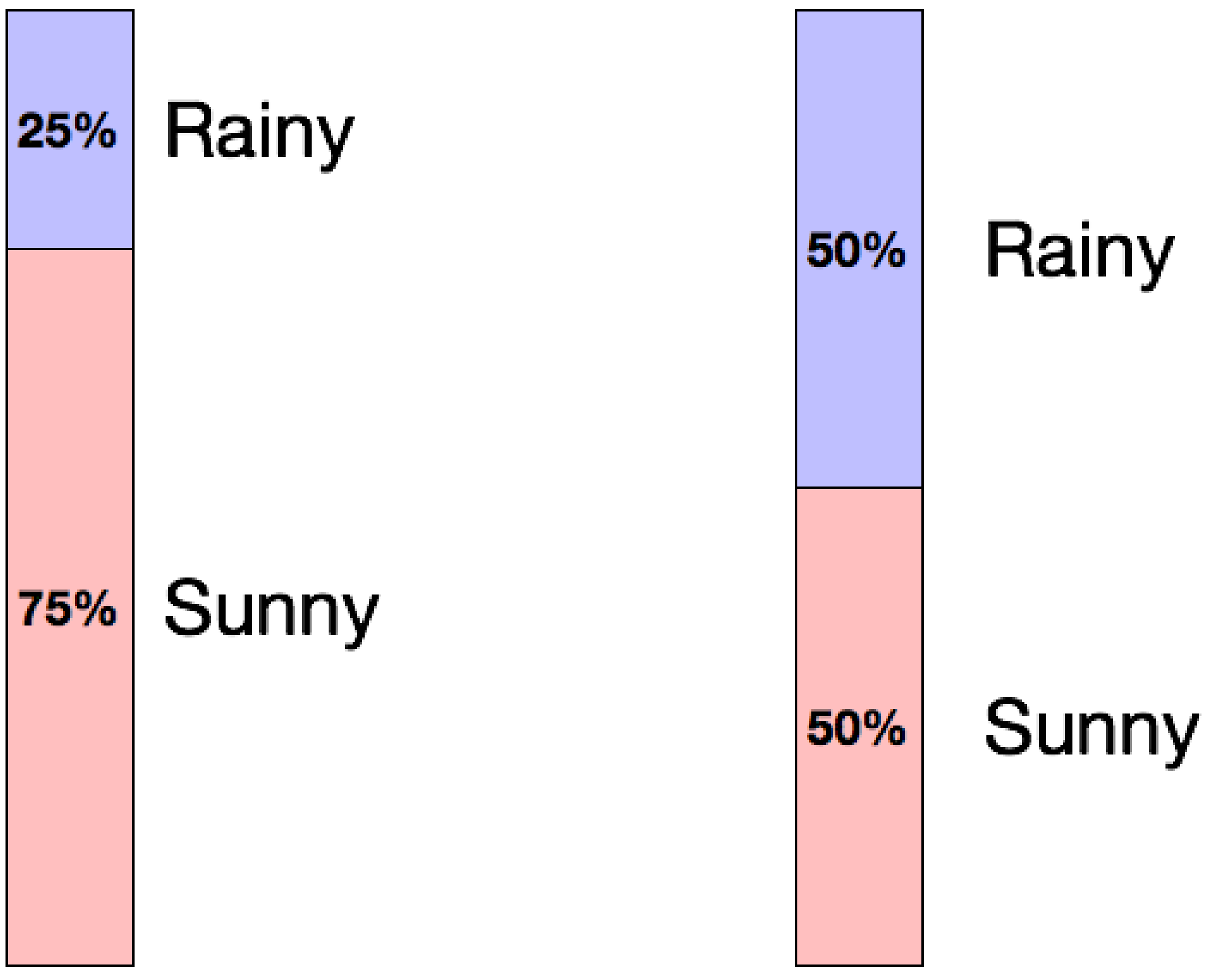
We used **6** characters

- Which one has a higher probability: T or H?
- Which one should carry more information: T or H?

# Motivation: Data compression

- Frequently occurring events should have short encodings
- We see this in English with words such as “a”, “the”, “and”, etc.
- We want to maximize the information-per-character
- Seeing common events provides little information
- Seeing uncommon events provides a lot of information

# Uncertainty and Information




**Which day is more uncertain?**

**How do we quantify uncertainty?**

Less likely → More uncertainty → More information → High entropy

# Recap

- Prob vs. Likelihood

dist. given

$\theta$ given

$P(\text{data} \mid \theta)$

data given

$P(\theta \mid \text{data})$

- MLE

This likely follows gaussian. a, b

maximized the likelihood of observed data points

$$L(\theta \mid x) = \prod_{i=1}^{n} f(x^{\{i\}} \mid \theta)$$

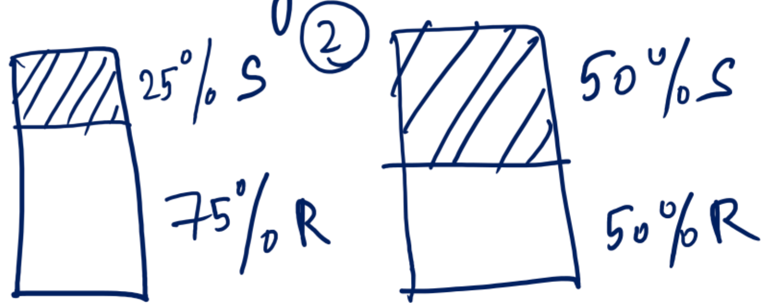
$$\log L(\theta \mid x) = \ell(\theta \mid x) = \sum_{i=1}^{n} \log\left(f(x^{\{i\}} \mid \theta)\right)$$

max. $\dfrac{\partial \ell}{\partial \theta} = 0 \longrightarrow$ optimized $\theta$

- Information

more uncertainty $\longrightarrow$ more information $\longrightarrow$ random variables

① ②




$I(x = S)$ will be higher case ①

$I(x = R)$ will be lower case ①

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# Entropy

- Average amount of information to encode a random variable $X$ with respect to its distribution $p(x)$ is the entropy $H(x)$:

$$E[g(x)] = \sum_x g(x)p(x)$$

Entropy

expected value of Information

$$H(x) = E[I(x)] = \sum_x I(x)p(x) = \sum_x p(x)\log_2\left(\frac{1}{p(x)}\right) = \boxed{-\sum_x p(x)\log_2 p(x)}$$

Considering a random variable $X$ with $k$ possible states:

$$H(x) = -\sum_{k=1}^{K} p(x=k)\log_2 p(x=k) = \sum_{k=1}^{K} p(x=k)\log_2\frac{1}{p(x=k)}$$

- Entropy is the **average amount of surprise (information)** you gain when observing outcomes of $X$.



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# Entropy

- Max possible entropy a random variable $X$ with $k$ possible states is $\log_2 k$

Coin flip

Unbiased coin flip

$$H(x) = \sum_x I(x)\, p(x)$$

$$= I(x=H)\, p(x=H) + I(x=T)\, p(x=T)$$

$$= \frac{1}{2} \log_2 \frac{1}{1/2} + \frac{1}{2} \log_2 \frac{1}{1/2}$$

$$= \log_2 \frac{1}{1/2} = \log_2 2$$

k=2

$$\underbrace{\text{entropy}}_{X\text{-ve}} = \sum \underbrace{\text{information}}_{X\text{-ve}} \times \underbrace{\text{probability}}_{X\text{-ve}}$$

Die Toss

Unbiased die

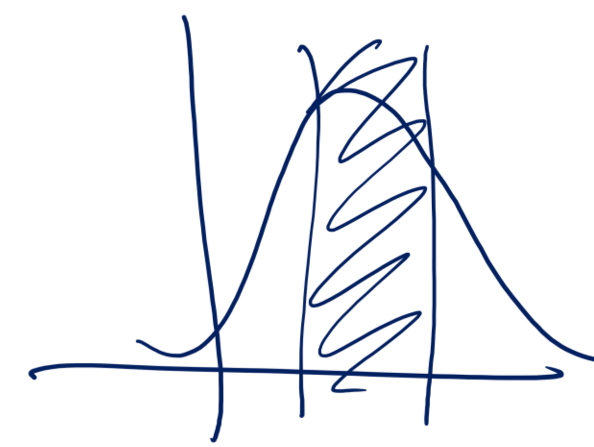
$$H(x) = I(x=1)\, p(x=1) + I(x=2)\, p(x=2) + \dots + I(x=6)\, p(x=6)$$

$$= \frac{1}{6} \log_2 \frac{1}{1/6} \times 6$$

$$= \log_2 6$$

k=6

Pmf



more uncertain

higher entropy

- Entropy is non-negative

- Most efficient code assigns $\boxed{-\log_2 P(x=k)}$ bits to encode the message $x = k$.
  - because it matches the information content of that event.

Information for r.v. X

# Example: entropy computation for coin toss

$\sum_x I(x)\,p(x)$

$\log_2 \frac{1}{p(x)}$

$$H(S) \equiv -(p_+ \log_2 p_+ + p_- \log_2 p_-)$$

biased

| | |
|---|---|
| Head | 0 |
| Tail | 6 |

$$p(H) = \frac{0}{6} = 0, \qquad p(T) = \frac{6}{6} = 1$$
$$H = -0\log_2 0 - 1\log_2 1 = 0$$

| | |
|---|---|
| Head | 1 |
| Tail | 5 |

$$p(H) = \frac{1}{6}, \qquad p(T) = \frac{5}{6}$$
$$H = -\frac{1}{6}\log_2\frac{1}{6} - 5/6\log_2\frac{5}{6} = 0.65$$

| | |
|---|---|
| Head | 2 |
| Tail | 4 |

$$p(H) = \frac{2}{6}, \qquad p(T) = \frac{4}{6}$$
$$H = -\frac{2}{6}\log_2\frac{2}{6} - 4/6\log_2\frac{4}{6} = 0.92$$

unbiased

| | |
|---|---|
| Head | 3 |
| Tail | 3 |

$$p(H) = \frac{1}{2}, \qquad p(T) = \frac{1}{2}$$
$$H = -\frac{1}{2}\log_2\frac{1}{2} - 1/2\log_2\frac{1}{2} = 1 \quad \log_2 K \quad \log_2 2 = 1$$



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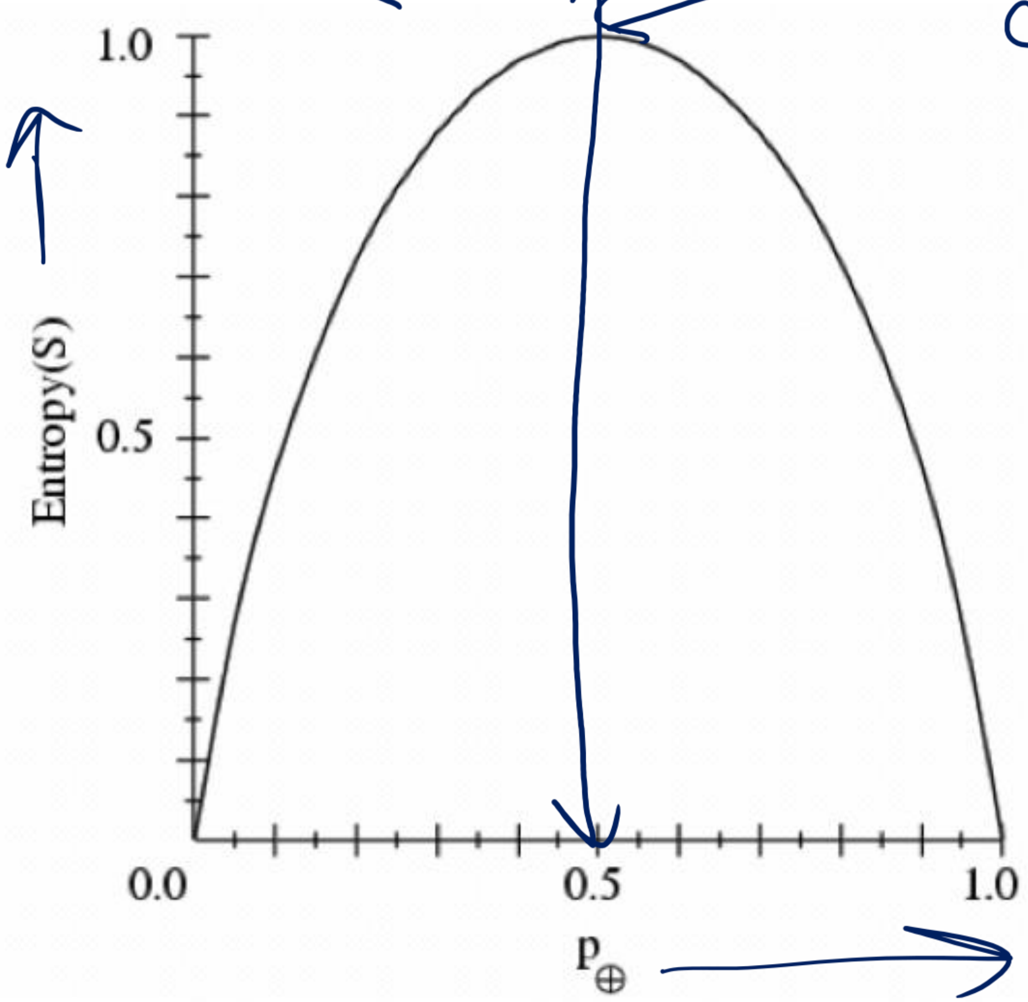
# Example: entropy

- $S$ is a sample of coin flips
- $p_+$ is the proportion of heads in $S$
- $p_-$ is the proportion of tails in $S$
- Entropy measures the uncertainty of $S$

$$H(S) \equiv -(p_+ \log_2 p_+ + p_- \log_2 p_-)$$

0,6 ← 1,5 2,4 3,2 2,3 → 4,2, 5,1, 6,0

unbiased case




prob.

# Uncertainty and Information

$E[x] =$

$-\log_2(p(x=rainy))\,p(x=rainy)$

$-\log_2(p(x=sunny))\,p(x=sunny)$

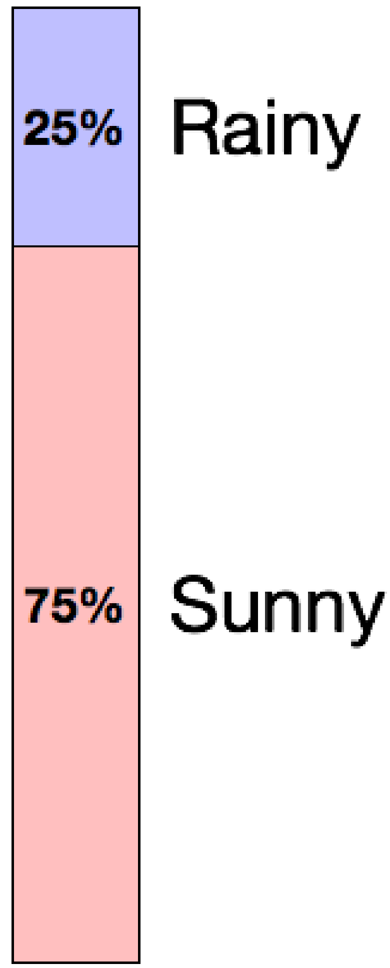
Case 1

$-\log_2 \frac{1}{4} \cdot \frac{1}{4} - \log_2 \frac{3}{4} \cdot \frac{3}{4}$

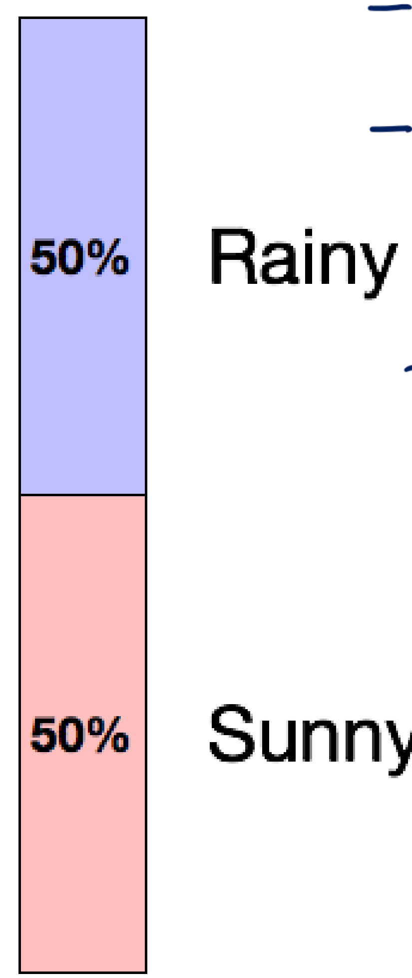
$\frac{1}{4} \log_2 4 + \frac{3}{4} \log_2 4/3$

$= 0.8$

Case 1




Case 2




Case 2

$-\log_2(p(x=rainy))\,p(x=rainy)$

$-\log_2(p(x=sunny)) \cdot p(x=sunny)$

$-\log_2 \frac{1}{2} \cdot \frac{1}{2} - \log_2 \frac{1}{2} \cdot \frac{1}{2}$

$\frac{1}{2} \log_2 2 + \frac{1}{2} \log_2 2$

$= \log_2 2$

$= 1$

**Which day is more uncertain?**

**How do we quantify uncertainty?**

Less likely → More uncertainty → More information → High entropy system

# Properties of Entropy

Coin toss → 1 bit
Die toss → 2.58 bits

- Non-negative: $H(P) \geq 0$
- Invariant with respect to permutation of its inputs:

$$H(p_1, p_2, \ldots, p_k) = H\left(p_{\tau(1)}, p_{\tau(2)}, \ldots, p_{\tau(k)}\right)$$

- For any other probability distribution $\{q_1, q_2, \ldots, q_k\}$

$$H(P) = \sum_i p_i \log \frac{1}{p_i} < \sum_i p_i \log \frac{1}{q_i}$$

Cross Entropy

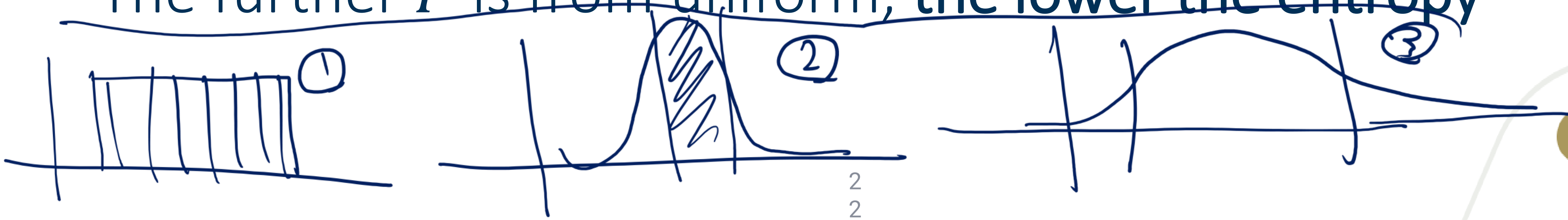
actual pdf

predicted pdf

- $H(P) \leq \log_2 \mathrm{k}$, with equality iff $p_i = \frac{1}{\mathrm{k}}, \forall i$

Entropy
1 > 3 > 2

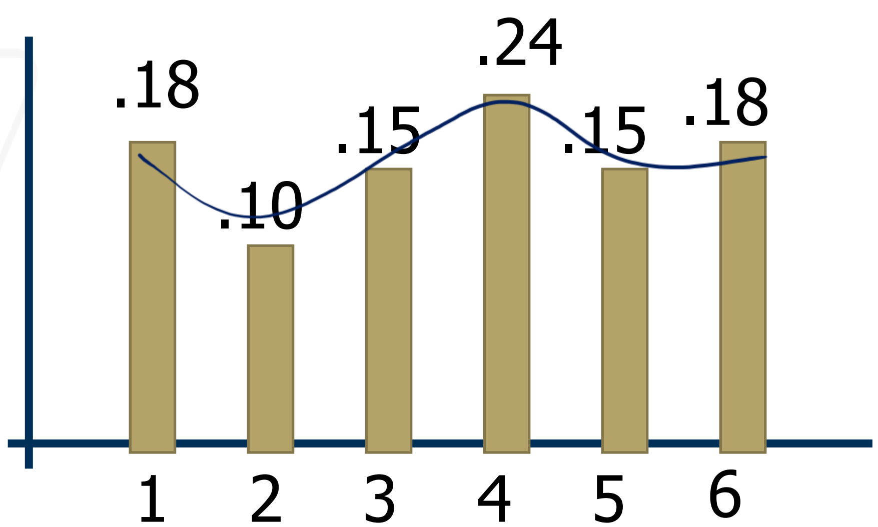
- The further $P$ is from uniform, **the lower the entropy**



# Example: entropy for rolling a die

$$H(x) = -\sum_{k=1}^{K} p(x=k)\log_2 p(x=k)$$

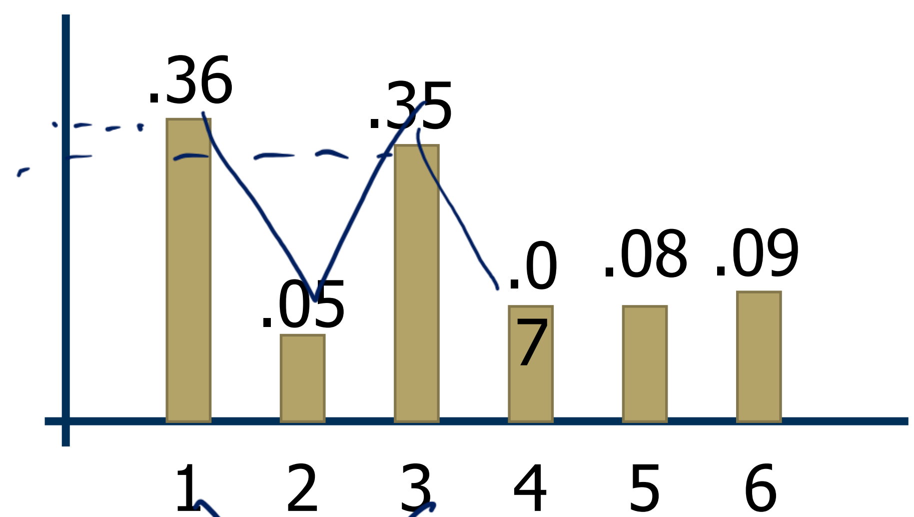
1




less biased

more entropy

2




more likely

# In context: Why does Entropy matter?

**Entropy measures uncertainty.**

- It tells us how much surprise to expect on average.
- High entropy = we should expect to learn more once the outcome is revealed.

**Entropy helps us compare systems**

- Coin that always lands heads → entropy = 0 (no uncertainty, no info).
- Fair coin → entropy = 1 bit (max for binary outcomes).
- Six-sided die → entropy ≈ 2.58 bits (more uncertainty than a coin).

👉 **Why helpful?**

- It lets us compare "information richness" between random processes.
- Used in feature selection. Features with higher entropy have more capacity to carry information.

**Big picture intuition**

- Entropy matters because it tells us how much "space" we need to represent uncertainty.
- The more uncertain the world is, the more information we gain by observing it — and entropy quantifies that.

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# Outline

- Motivation
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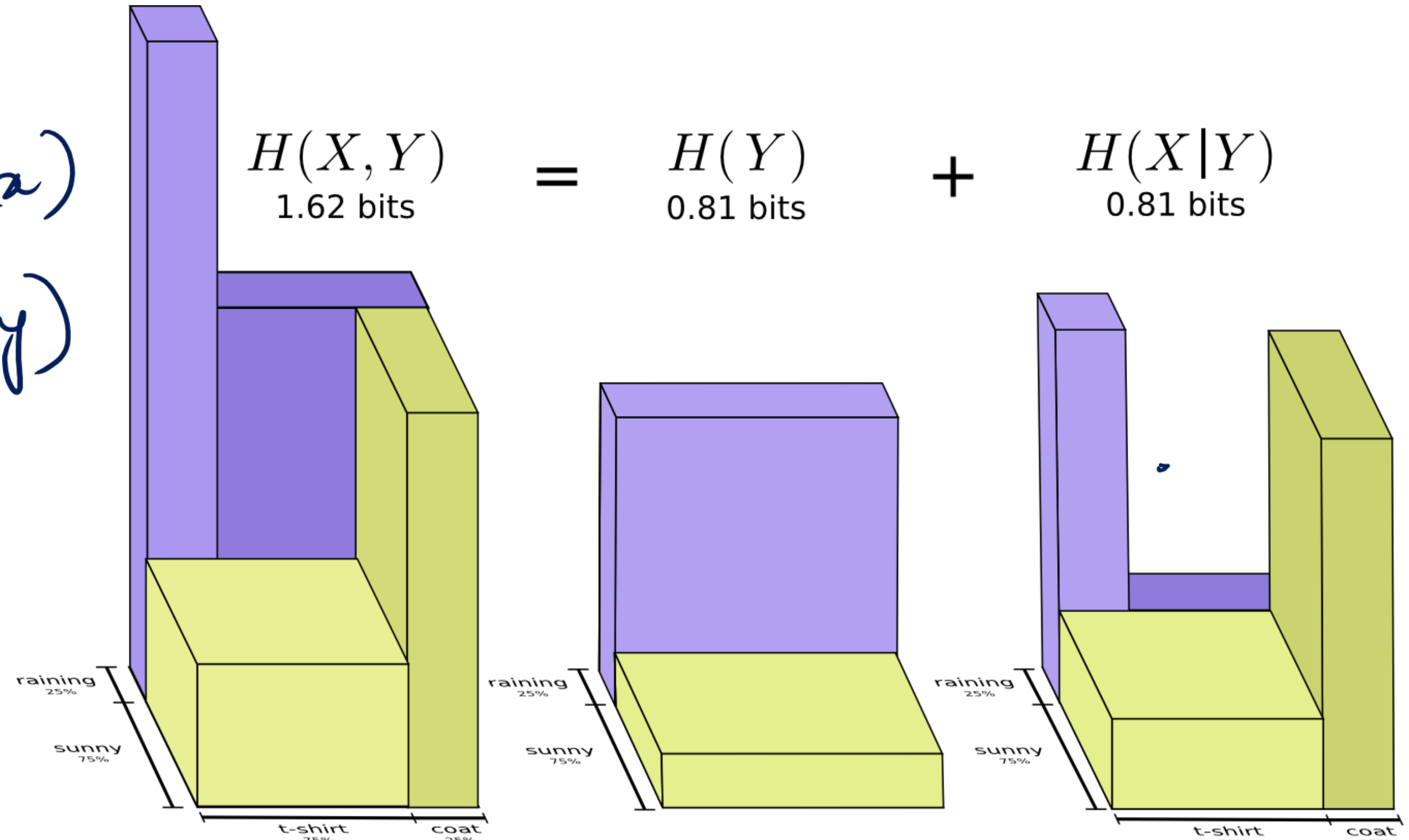
- Conditional Entropy and Mutual Information
- Cross-Entropy and KL-Divergence

# Entropy

Do not like list

$$H = \sum p(x) \log_2 \frac{1}{p(x)}$$

$$p(x, y) = p(y \mid x)\, p(x)$$
$$= p(x \mid y)\, p(y)$$




$$\underline{H(x,y)} = \underline{H(y \mid x)} + H(x)$$

log in the definition

Joint Entropy

Conditional Entropy

# Joint Entropy

$H(x) = \sum_{x} p(x) \log_2 \frac{1}{p(x)}$

- The joint entropy measures the entropy of a joint probability distribution of two random variables $X$ and $Y$

$$H(X,Y) = \sum_{x \in X, y \in Y} p(x_i, y_i) \log \frac{1}{p(x_i, y_i)}$$

- It helps us quantify the **dependence** between different variables and how much information they share

- From the product rule we can also see that

$$H(X,Y) = H(X) + H(Y|X)$$

# Joint Entropy

$P(T = \text{mild}, M = \text{high}) = 0.1$

$H(T) = \sum_T p(T) \cdot \log_2 \frac{1}{p(T)} = 0.3 \log_2 \frac{1}{0.3} + 0.5 \log_2 \frac{1}{0.5} + 0.2 \log_2 \frac{1}{0.2}$

$H(T) + H(M) = 2.456 \geq H(T,M)$

$H(T,M) = 2.32$

$p(T=\text{cold} \mid M=\text{low})$

$\frac{0.1}{0.6} = \frac{1}{6}$

$p(x,y) = p(x) \cdot p(y)$

$x \perp y$

$H(x,y) = H(x) + H(y)$

$x \perp y$

Temperature

huMidity

| | cold | mild | hot | |
|---|---|---|---|---|
| low | 0.1 | 0.4 | 0.1 | 0.6 |
| high | 0.2 | 0.1 | 0.1 | 0.4 |
| | 0.3 | 0.5 | 0.2 | 1.0 |

- $H(T) = H(0.3, 0.5, 0.2) = 1.48548$
- $H(M) = H(0.6, 0.4) = 0.970951$
- $H(T) + H(M) = 2.456431$
- **Joint Entropy**: consider the space of $(t, m)$ events $H(T, M) = \sum_{t,m} P(T = t, M = m) \cdot \log \frac{1}{P(T=t, M=m)}$
  $H(0.1, 0.4, 0.1, 0.2, 0.1, 0.1) = 2.32193$

$\sum_M p(M) \log_2 \frac{1}{p(M)} = 0.6 \log_2 \frac{1}{0.6} + 0.4 \log_2 \frac{1}{0.4}$

$H(T,M) = \sum_{T,M} p(T,M) \log_2 \frac{1}{p(T,M)}$

6 different terms

Notice that $H(T, M) \leqslant H(T) + H(M)$ !!!

$$H(T,M) = H(T|M) + H(M) = H(M|T) + H(T)$$

# Conditional Entropy

$H(T) = 1.48$
$H(T|M) = 1.35$

$H(x) \geq H(x|y)$

$x \perp y \rightarrow$ equality

$$H(Y|X) = \sum_{x \in X} p(x) H(Y|X=x) = \boxed{\sum_{x \in X, y \in Y} p(x,y) log \frac{p(x)}{p(x,y)}}$$

$$H(Y|X=x) = \sum_{y} p(Y|X=x) \cdot \log_2 \frac{1}{p(Y|X=x)}$$

$H(T|M=low) =$ sum over all temperatures

$p(T=cold|M=low) \cdot \log_2 \frac{1}{p(T=cold|M=low)}$

$+ p(T=mild|M=low) \cdot \log_2 \frac{1}{p(T=m|M=l)}$

$+ p(T=hot|M=low) \cdot \log_2 \frac{1}{p(T=h|M=l)}$

$$P(T = t|M = m)$$

| | cold | mild | hot | |
|---|---|---|---|---|
| low | 1/6 | 4/6 | 1/6 | 1.0 |
| high | 2/4 | 1/4 | 1/4 | 1.0 |

**Conditional Entropy**:

- $H(T|M = low) = H(1/6, 4/6, 1/6) = 1.25163$ $= \frac{1}{6} \log_2 \frac{1}{1/6} + \frac{4}{6} \log_2 \frac{1}{4/6} + 1/6 \log_2 \frac{1}{1/6}$
- $H(T|M = high) = H(2/4, 1/4, 1/4) = 1.5$
- **Average Conditional Entropy** (aka equivocation):
  $H(T/M) = \sum_m P(M = m) \cdot H(T|M = m) =$
  $0.6 \cdot H(T|M = low) + 0.4 \cdot H(T|M = high) = 1.350978$

# Conditional Entropy

$$P(M = m|T = t)$$

| | cold | mild | hot |
|---|---|---|---|
| low | 1/3 | 4/5 | 1/2 |
| high | 2/3 | 1/5 | 1/2 |
| | 1.0 | 1.0 | 1.0 |

Conditional Entropy:

- $H(M|T = cold) = H(1/3, 2/3) = 0.918296$
- $H(M|T = mild) = H(4/5, 1/5) = 0.721928$
- $H(M|T = hot) = H(1/2, 1/2) = 1.0$
- Average Conditional Entropy (aka Equivocation): $H(M/T) = \sum_t P(T = t) \cdot H(M|T = t) =$ $0.3 \cdot H(M|T = cold) + 0.5 \cdot H(M|T = mild) + 0.2 \cdot H(M|T = hot) = 0.8364528$

# Conditional Entropy

- Conditional entropy $H(Y|X)$ of a random variable $Y$ given $X_i$

Discrete random variables:

$$H(Y|X) = \sum_{x \in X} p(x_i) H(Y|X = x_i) = \sum_{x \in X, y \in Y} p(x_i, y_i) log \frac{p(x_i)}{p(x_i, y_i)}$$

Mixed setting: Continuous (over x) and discrete (over y):

$$H(Y|X) = -\int \left( \sum_{k=1}^{K} p(y = k|x_i) \log_2 (y = k|x_i) \right) \mathrm{p(x_i)dx_i}$$

# Mutual Information

$H(Y) \geq H(Y|X)$

$X \; [\text{ht.} \;\; \text{wt.} \;\; \text{eye.} \;\; \text{fur.} \;\; \cdots] = Y \; [\text{cat}, \text{dog}, \text{dog}, \vdots]$

- Mutual information: quantify the reduction in uncertainty in $Y$ after seeing feature $X_i$

classification label

$$\boxed{I(X_i, Y) = H(Y) - H(Y|X_i)}$$

feature

- The more the reduction in entropy, the more informative a feature.

- Mutual information is symmetric
  - $I(X_i, Y) = I(Y, X_i) = H(X_i) - H(X_i|Y)$
  - $I(Y|X) \; = \int \sum_k^K p(x_i, y = k) \log_2 \frac{p(x_i, y=k)}{p(x_i)p(y=k)} \; dx_i$
  - $= \int \sum_k^K p(x_i|y = k) p(y = k) \log_2 \frac{p(x_i|y = k)}{p(x_i)} \; dx_i$

$H(X) \geq H(X|Y)$

# Properties of Mutual Information

$$I(X,Y) = H(X) - H(X|Y)$$

non negative

$$= \sum_x P(x) \cdot \log \frac{1}{P(x)} - \sum_{x,y} P(x,y) \cdot \log \frac{1}{P(x|y)}$$

$$= \sum_{x,y} P(x,y) \cdot \log \frac{P(x|y)}{P(x)}$$

$$= \sum_{x,y} P(x,y) \cdot \log \frac{P(x,y)}{P(x)P(y)}$$

Properties of Average Mutual Information:

- Symmetric
- Non-negative
- Zero iff $X, Y$ independent

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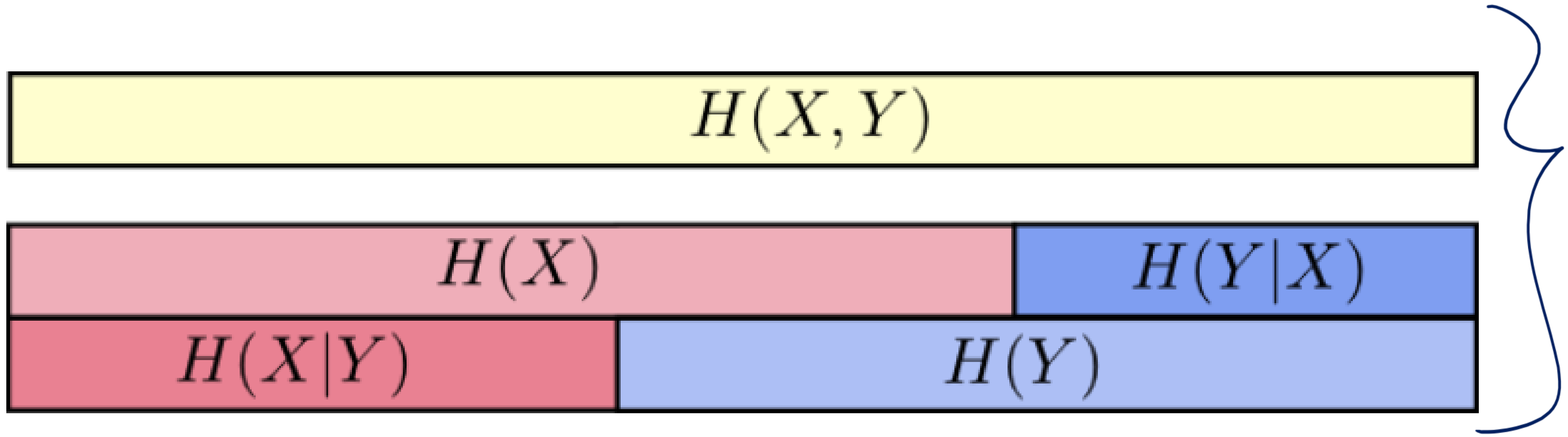
# CE and MI: Visual Illustration

$$H(x,y) = H(x|y) + H(y)$$
$$= H(y|x) + H(x)$$



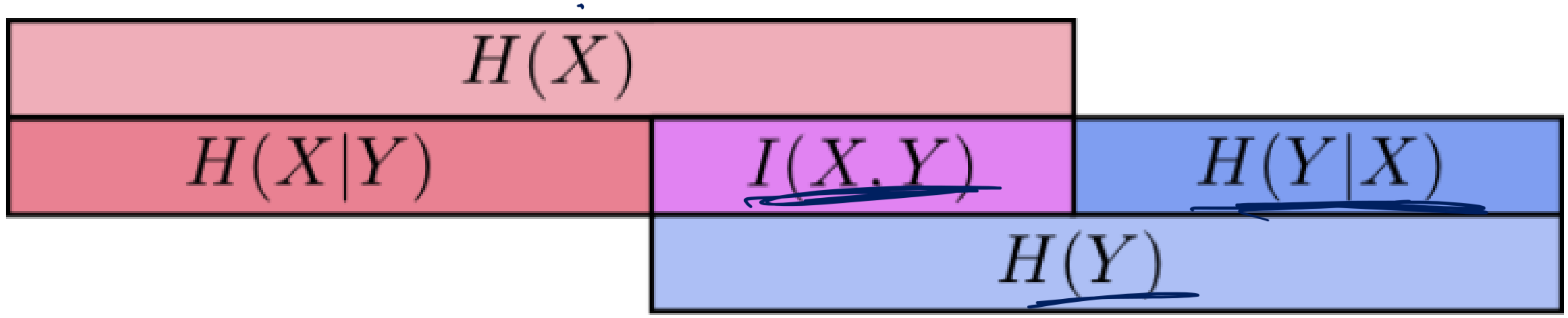




Image Credit: Christopher Olah.

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We will cover this next class in our Optimization lecture



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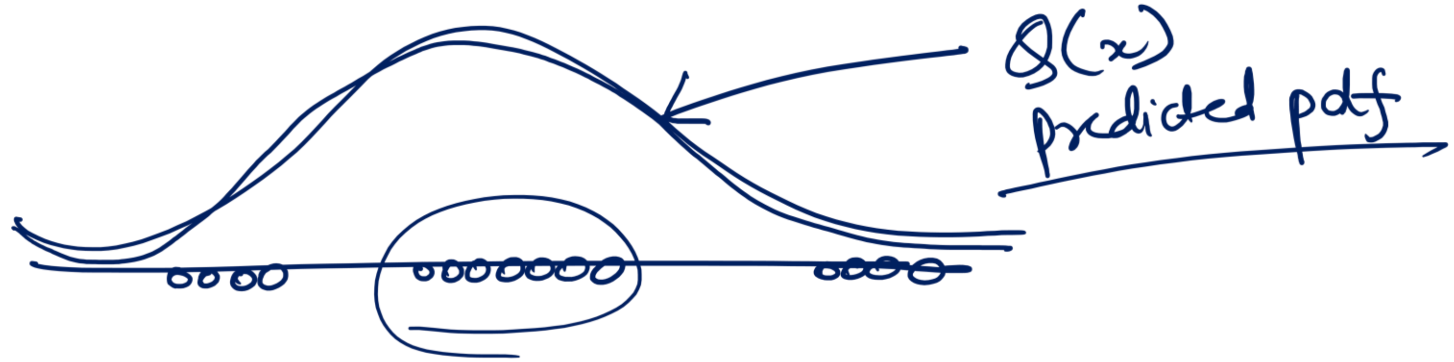
# Cross Entropy




Suppose the **true distribution** of data is $P(x)$.

But you assume or model the data using another distribution $Q(x)$.

**Cross entropy** measures the *average number of bits required* to encode samples from $P$, **if you use a code optimized for** $Q$ **instead of** $P$.

👉 In short: *How inefficient is your code when you use the wrong distribution?*

# Entropy Summary

- Entropy (H(X)) is the average number of bits needed to encode the random variable $X$ when using an optimal code matched to its distribution $p(x)$.

- Joint Entropy (H(X,Y)) is the average number of bits needed to encode both $X$ and $Y$ together, if you use an optimal code matched to their joint distribution.

- Conditional Entropy (H(Y|X)) is the average number of extra bits needed to encode $Y$, once you already know $X$.

- Cross Entropy is the average number of bits needed to encode data that really follows distribution $P$, if you instead use a code optimized for another distribution $Q$.

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# Cross Entropy

H(p, q) is the Cross Entropy (CE)

**Cross Entropy**: The expected number of bits when a wrong distribution Q is assumed while the data actually follows a distribution P

KL Divergence

$$\underbrace{H(p,q)}\ = -\sum_{x\in\mathcal{X}} p(x)\,\log q(x) = H(P) + \boxed{KL[P][Q]}$$

p(x) → true dist.

Q(x) → predicted dist. that we are using to model the data

H(P): the true entropy (best-case cost, if you used the right distribution).

$KL$[P][Q]: the extra penalty (inefficiency) you pay for using the wrong distribution $Q$.

# Cross Entropy

**Cross Entropy**: The expected number of bits when a wrong distribution Q is assumed while the data actually follows a distribution P

$$H(p,q) = -\sum_{x\in\mathcal{X}} p(x)\, \log q(x) = H(P) + KL[P][Q]$$

This is because:

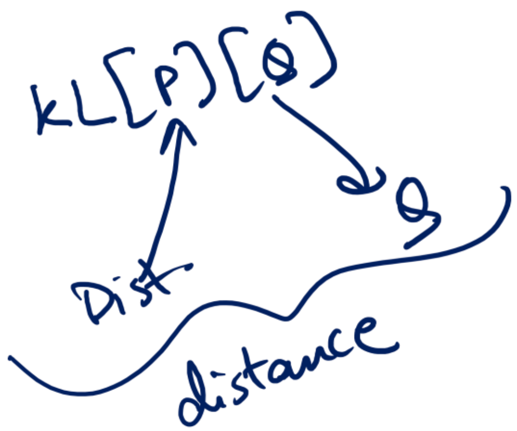
$$H(p,q) = \mathrm{E}_p[l_i] = \mathrm{E}_p\left[\log \frac{1}{q(x_i)}\right]$$

$$H(p,q) = \sum_{x_i} p(x_i)\, \log \frac{1}{q(x_i)}$$

$$H(p,q) = -\sum_{x} p(x)\, \log q(x).$$

# Kullback-Leibler Divergence

Another useful information theoretic quantity measures the difference between two distributions.




$$\textbf{KL}[P(S)\|Q(S)] = \sum_s P(s) \log \frac{P(s)}{Q(s)}$$

$$= \underbrace{\sum_s P(s) \log \frac{1}{Q(s)}}_{\text{cross entropy}} - \mathbf{H}[P] = H(P,Q) - H(P)$$

KL Divergence is a **KIND OF** distance measurement

Excess cost in bits paid by encoding according to $Q$ instead of $P$.

$$-\textbf{KL}[P\|Q] = \sum_s P(s) \log \frac{Q(s)}{P(s)}$$

log function is concave or convex?

$$\sum_s P(s) \log \frac{Q(s)}{P(s)} \leq \log \sum_s P(s) \frac{Q(s)}{P(s)}$$

By Jensen Inequality

$$= \log \sum_s Q(s) = \log 1 = 0$$

So $\textbf{KL}[P\|Q] \geq 0$. Equality iff $P = Q$

When $P = Q,\ KL[P||Q] = 0$

# Take-Home Messages

- Entropy
  - A measure for uncertainty
  - High uncertainty maps to High information and High entropy
  - Why it is defined in this way (optimal coding)
  - Its properties
- Joint Entropy, Conditional Entropy, Mutual Information
  - The physical intuitions behind their definitions
  - The relationships between them
- Cross Entropy, KL Divergence
  - The physical intuitions behind them
  - The relationships between entropy, cross-entropy, and KL divergence